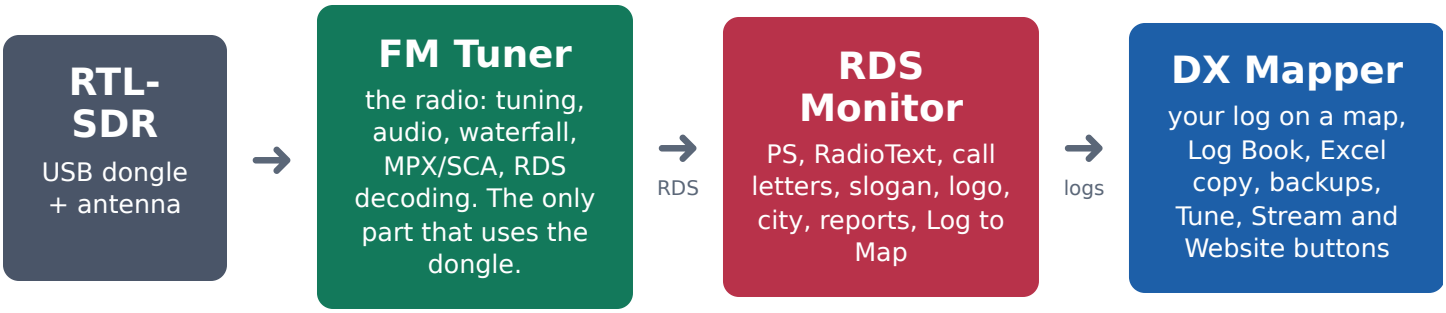
DXMapper Suite — Quick Start

One radio program in three parts: **FM Tuner**, **RDS Monitor** and **DX Mapper**. For DXing and for everyday listening.



↶ Tune buttons in the RDS Monitor and DX Mapper tune the FM Tuner

Install (first time only)

1. **Install Python 3.10 or newer (64-bit)** from python.org. Tick *"Add python.exe to PATH"*.
2. **Extract the ZIP** into `C:\`. It holds one folder, **DXMapper Suite** — one program in one folder. In Windows "Extract All", change the path to `C:\` so you don't get the folder twice.
3. **Double-click "Install All.bat".** It installs what all three parts need and tells you if anything is missing.
4. **FM Tuner only:** run **Zadig** once (driver: WinUSB) and copy `rtlsdr.dll` + `libusb-1.0.dll` into the main folder. Install All.bat checks this; INSTALL.txt shows how, step by step.
5. **HackRF (SoapySDR) support is EXPERIMENTAL** until it is fixed. Use an RTL-SDR dongle for now.

```
C:\DXMapper Suite\
  Start Radio.pyw   every day: opens all three
  FM_Tuner.py       quick launch: just the radio
  RDS Monitor.py    quick launch: just the RDS display
  DX Mapper.py      quick launch: just the map log
  settings\         all your data
  QUICK_START.pdf   this guide
  WHATS_NEW.txt     what changed in each update, newest first
  OWNERS_MANUAL.pdf every screen and button in detail
  Install All.bat   first time / after Python updates
```

From an older copy (e.g. `C:\Radio`): rename its folder to **DXMapper Suite** first, then extract to `C:\`. On first start your old settings and rtlsdr.dll are **copied** into the one settings folder; the old folders are left alone (delete them once it works).

Every day

1. **Double-click “Start Radio.pyw”.** It opens the FM Tuner, then the RDS Monitor, then DX Mapper — no black console windows.
2. The RDS Monitor connects to the tuner by itself (if not: *Input › FM Tuner (TCP)*).
3. Tune in the FM Tuner, or press **Tune** next to any station in DX Mapper.

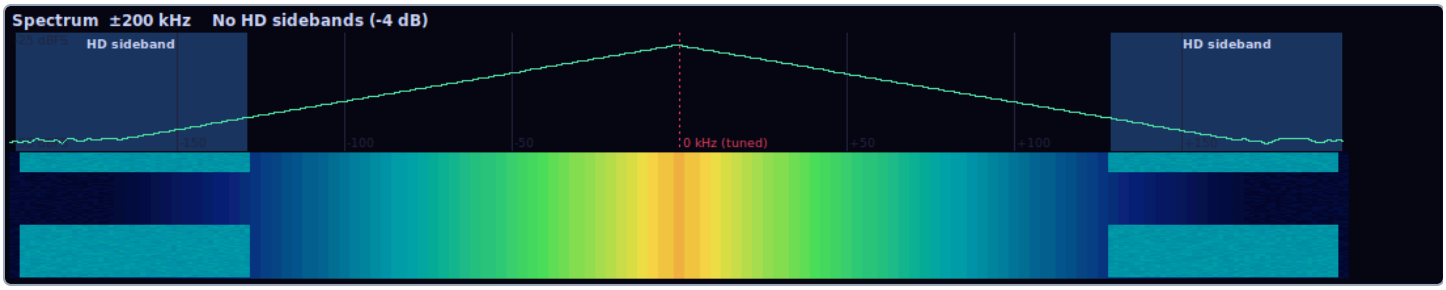
FM Tuner — the radio



FM Tuner window, opened without a dongle (so no signal or RDS is showing).

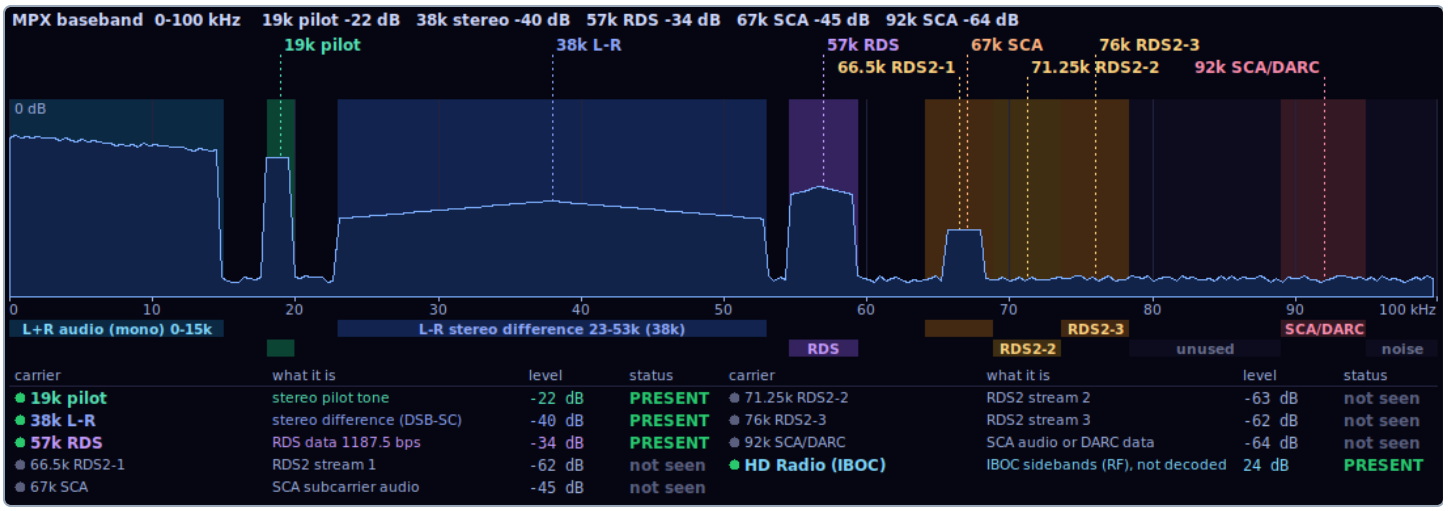
1. **Frequency** — click and type a new one
2. **Step ±0.2** and **Seek** up/down
3. **Presets** — save, rename, delete; one click tunes
4. **Live RDS** — PI and PS as decoded
5. **Record MPX** — a sample of the signal for later
6. **View** menu — show/hide panels: waterfall, MPX, SCA, DX filter ...

Waterfall — is this station HD?



RF spectrum and waterfall. HD Radio shows as blocks either side of the station (shaded). Built-in demo picture with a made-up signal.

MPX baseband — what the station is sending



Stereo pilot, L−R, RDS, SCA and RDS2, each labelled with its level and PRESENT / not seen. Built-in demo picture with a made-up signal.

RDS Monitor — what the station is sending



RDS Monitor replaying a European test recording. Its PI code doesn't spell out call letters, so it shows (unknown) until the station is named in DX Mapper.

1 **Input** menu — FM Tuner, a .spy file, or another RDS source (with the FM Tuner, a Tuned frequency row with Tune/Seek appears under the title)

2 **Call letters, logo, slogan, city** — from DX Mapper's database

3 **PI, PTY and PS** as decoded

4 **Signal quality** — how much of the RDS was decoded

5 **Decode badges** — green = complete, yellow = partly received

6 **TMC traffic ticker** — scrolls along the bottom when a station sends traffic messages

7 **Tools** — Log to DX Mapper (Ctrl+L), Auto-log, Home/Away location, reports

Automatic call letters are US only. The monitor works out call letters from the PI code only for US stations. Outside the US (Canada, Mexico, Europe…) name the station once in DX Mapper (click it, then **Edit**); the monitor shows it from then on.

Logging a catch

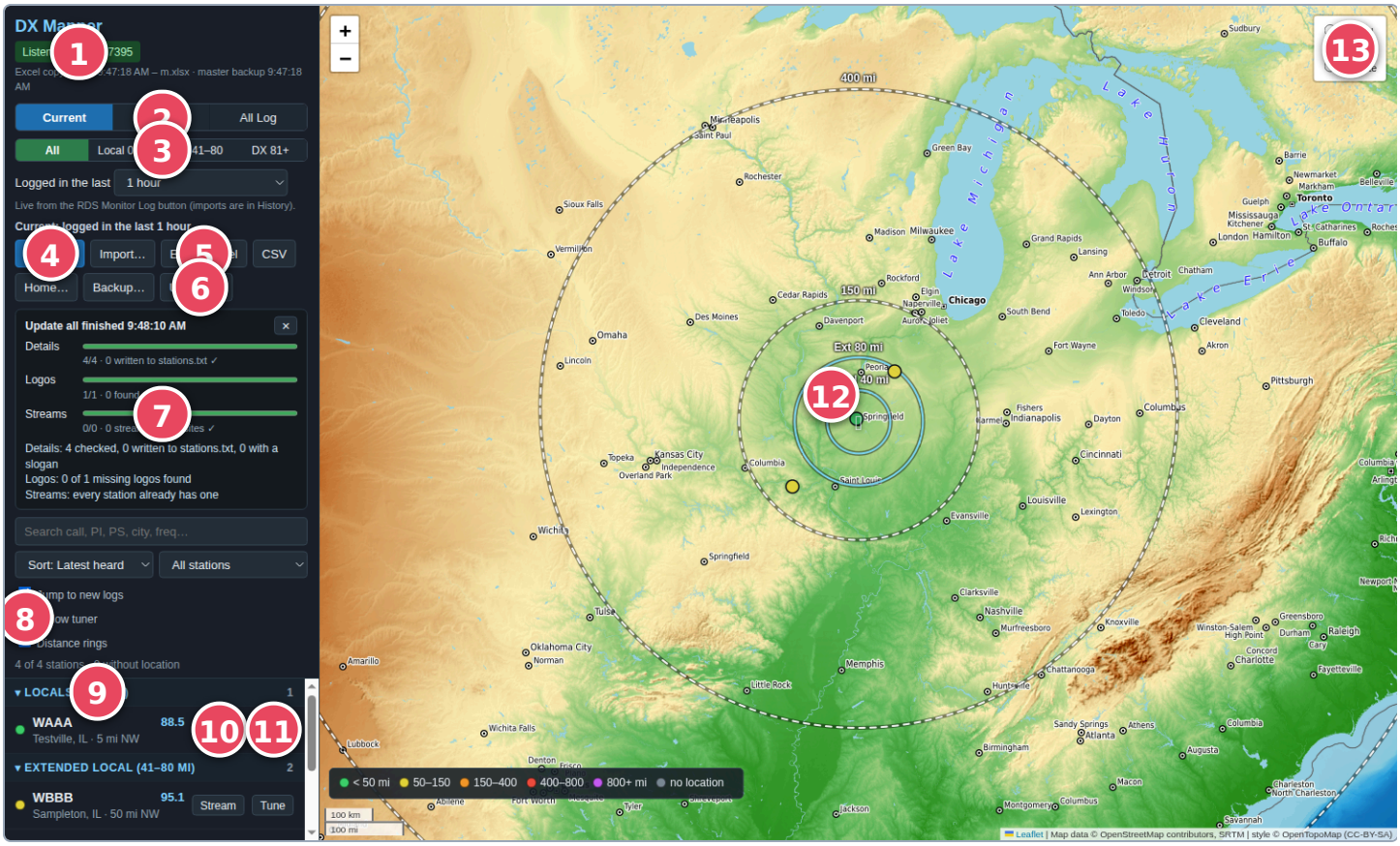
1. Press **Ctrl+L** (or *Tools › Log Current Station to DX Mapper*). The station goes onto the map with its full RDS report.

2. Or turn on *Tools › Auto-log to DX Mapper* (off by default). It logs once enough RDS is decoded and keeps updating that log while you stay on the station. Nothing decoded earlier is ever lost.

DXing away from home

Tools › Set Away Location… then *Use Away location*. Have a USB GPS dongle? *Tools › GPS Location…* sets it for you and can keep it updated as you travel (optional). DX Mapper's map and distances follow, and every log remembers where you were.

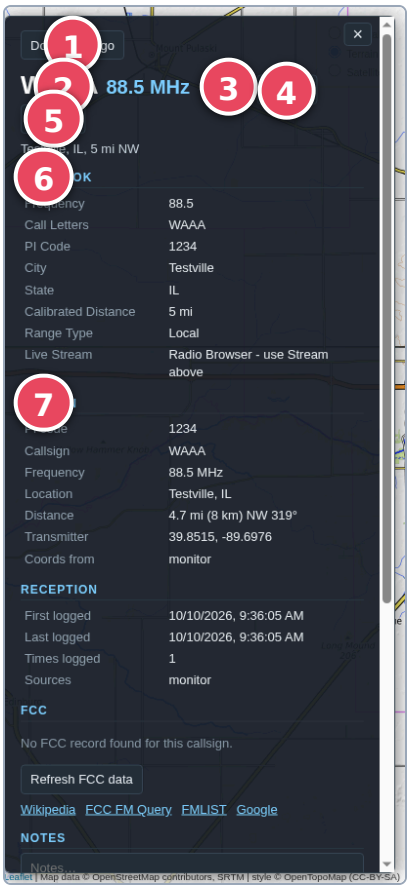
DX Mapper — your log on a map



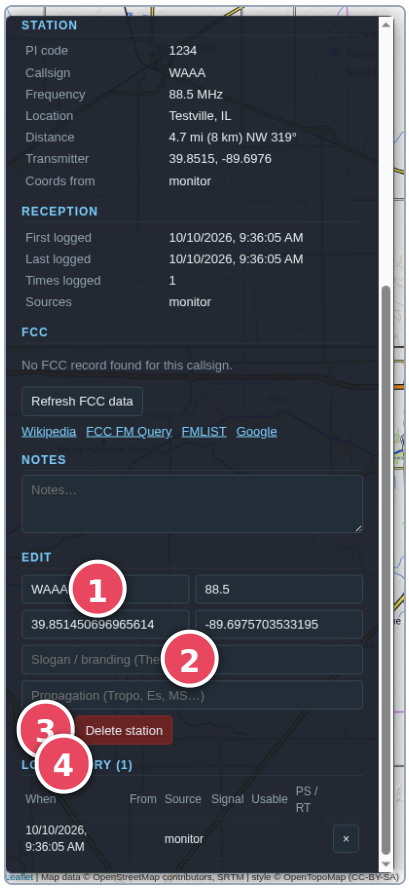
DX Mapper with made-up test stations and a made-up home location.

1. **Listening** for the RDS Monitor; **Now hearing** shows here
2. **Current / History / All Log** views
3. **All / Local / Ext / DX** filter
4. **Log Book** — your log as a table, one row per station
5. **Import, Export Excel, CSV, Home, Backup**
6. **Update all** — details, logos, streams and websites for your whole log
7. **Progress bars and summary** for Update all; errors in red
8. **Follow tuner** — the map follows whatever you tune
9. **Locals** (0–40 mi), **Extended local** (41–80) by frequency, then **DX**
10. **Stream** — the internet stream, in your browser
11. **Tune** — one click tunes the FM Tuner
12. **Markers** — colour = distance; click one for details
13. **Map type** — Normal, Terrain, Satellite

Station details & fixing a station



Click a marker or a station in the list.



Edit box at the bottom of the details.

1 **Logo** — or **Download logo** if there is none yet

2 **Call letters and MHz**

3 **Tune** — off the air

4 **Stream** — internet stream

5 **Website** — the station's own site

6 **Log Book** — your sheet's columns

7 **Station**, reception, FCC, notes and every log below

1 **Call letters** and MHz, then coordinates

2 **Slogan** — type your own

3 **Save**

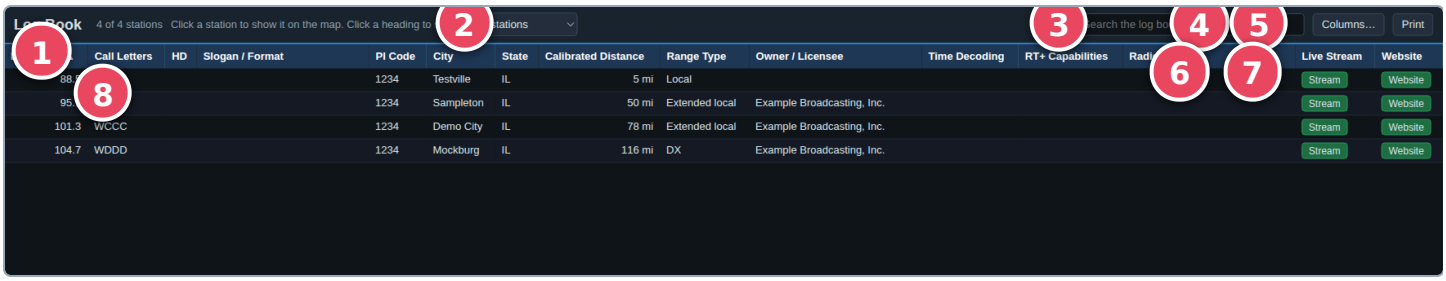
4 **Log history** — every catch, as it was caught

Fixing a station

1. Change **call letters, MHz, coordinates or slogan** in Edit and press **Save**.
2. New call letters look up their own location (FCC). The RDS Monitor reads DX Mapper's log, so it shows the fix within a second.
3. A slogan you type always wins over Wikipedia. Clear the box to remove it.

Update all (sidebar) goes through your whole log: fills the RDS Monitor's fallback stations.txt (call, city, coordinates, slogan), downloads missing logos, and looks up live streams and station websites. A box shows a progress bar for each step and a summary. Lines you typed and logos you have are kept.

Log Book — read your log like a log



Log Book with made-up test stations. Opens from the Log Book button in DX Mapper.

1. **Click any heading** to sort by it
2. **All / Locals / Extended locals / DX / HD** filter
3. **Search** any column
4. **Columns…** — pick which columns show
5. **Print** the log book
6. **Stream** — the live stream
7. **Website** — the station's site
8. **Click a station** and the map jumps to it

1 One row per station, with the columns of your spreadsheet: Frequency, Call Letters, HD, Slogan / Format, PI Code, City, State, Calibrated Distance, Range Type, Owner / Licensee, Time Decoding, RT+ Capabilities, RadioText, Live Stream and Website.

2 **Imported** columns go straight into each station. New catches fill in what is blank and never erase anything.

3 **HD stations** import as their own rows (`WKQX-HD2`, `101.1 HD2` or an HD column). An old HD catch keeps the name it had then; a new catch of it replaces the name with what you caught.

4 **Columns…** picks which columns show and print (saved in settings). **Export Excel** puts the Log Book, with every column, on the first sheet.

5 **Modern** (the Classic | Modern switch): station cards with Local / Extended Local / DX / Show all tabs and PI PTY RT RT+ iTunes TMC lights. Click a card for the station panel, Edit info and Delete station (asks first). It follows the map; **Open station card** in Station Details jumps to it. Regular stations are just call letters; HD2–HD8 show as `WAPD-HD2`.

6 **Doubles:** splatter auto logs join the station already logged; pressing **Log** in the RDS Monitor always logs where you're tuned. **Duplicates…** merges doubles already there (backup first). Full details: Owner's Manual 6.5.

Your data — the settings folder

```
settings\  one folder for all three:
  FM Tuner     presets, panel layout, gain memory
  RDS Monitor  stations.txt, logos, Home/Away, station counts
  DX Mapper    dx_log.db (first log), Logs\ (other logs), DX Log.xlsx, Backups
```

1. **Updating:** extract the new ZIP to the same place, over the old copy. The ZIP never contains the settings folder, so your presets, stations and log stay as they are.

2. **Backup:** DX Mapper › **Backup…** Put a Google Drive folder under “Also copy it to” for a copy off the PC. **Restore** brings it back after a reinstall.

3. **Excel:** settings\DX Log.xlsx is rewritten after every log — open it any time. **It is a copy for viewing: edits typed into it in Excel are NOT read back and get overwritten. Make changes in DX Mapper, or bring an edited sheet in with Import… (Excel or Google Sheets), which only fills blanks.**

4. **Several logs:** DX Mapper › **Logs…** Make Historical, Current or trip logs; each is its own file. Tick several to show together — the **first ticked** gets new catches, the rest are view only. **Copy to Current** brings an old station (and its HD channels) over. *Also log new catches into the other ticked logs* is **off by default** and asks first (Turn it on / Keep it off). Manual 6.8.

Thumb drive & travel laptop

The whole radio can run from a USB thumb drive. Set a laptop up once (Python, **Install All.bat** from the drive, Zadig), and after that you only carry the drive, the dongle and an antenna. Close all three programs before unplugging the drive. Full steps: Owner's Manual, chapter 2, *Running from a thumb drive*.

For experimenting

Each part also runs on its own; each folder has a **README.md** with every option and an **INSTALL.txt**. They talk to each other only on this PC (127.0.0.1):

Port	From → to	What
23	FM Tuner → RDS Spy & other RDS programs	RDS groups (ASCII G), the SDR# RDSOutput port.
7376	FM Tuner → RDS Monitor	Same RDS groups plus RDS2. Use it if port 23 is taken.
7373	FM Tuner → RDS Monitor	Telemetry: frequency, signal, stereo, HD
7374	anything → FM Tuner	Commands, e.g. `TUNE 101.1`
7375	RDS Monitor → DX Mapper	Logs and “Now hearing”
7380	DX Mapper	The map window / web page

The RDS Monitor also reads RDS Spy **.spy** files or any other ASCII G source (*Input* menu), so it works without the FM Tuner.

Screenshots use test recordings and test stations, not live reception.